Jaemin Park

+82 (010) 6718 3305 | arnoldpark03@gmail.com | arnold518.github.io/ | github.com/arnold518

EDUCATION

Seoul National University, *Dept. of Computer Science and Engineering* | Seoul, Korea — 2022 - Current
GPA: 3.47 / 4.3 Major GPA: 3.77 / 4.3

Gyeonggi Science High School | Suwon, Korea GPA: 4.03 / 4.3 — 2019 - 2021

Major Courses: : Computer Programming (OOP) | Principle of Programming (FP) | Automata Theory | Data Structure | Algorithm | Computer Architecture | Modern Algebra | Introduction to Mathematical Analysis

ACHIEVEMENTS

The 48th Annual World Finals of the International Collegiate Programming Contest - 9th Place (Bronze Medal) - Team *NewTrend* — 2024
The 2024 ICPC Asia Pacific Championship - 1st place (Champion) - Team *NewTrend* — 2024
The 2023 ICPC Asia Hue City Regional Contest - 1st place (Champion) - Team *NewTrend* — 2023
The 2024 ICPC Asia Seoul Regional Contest - 10th place - Team *aespa* — 2024
The 2023 ICPC Asia Seoul Regional Contest - 3nd place - Team *NewTrend* — 2023
The 2022 ICPC Asia Seoul Regional Contest - 4th place - Team *NewTrend* — 2022
Samsung Collegiate Programming Cup 2023 - 2nd Prize — 2023
Samsung Collegiate Programming Cup 2022 - 4th Prize — 2022
Korea Selection Test for International Olympiad of Informatics 2021 - 6th Place — 2021
Samsung Humantech Paper Award 2021 Mathematics / Data Processing section - Gold Prize — 2021
Samsung Humantech Paper Award 2020 Mathematics / Data Processing section - Bronze Prize — 2020
Korean Olympiad of Informatics 2021 - Silver Prize (6th Place) — 2021
Korean Olympiad of Informatics 2020 - Silver Prize (5th Place) — 2020
Korean Olympiad of Informatics 2019 - Silver Prize — 2019
Nexon Youth Programming Challenge 2021 - Bronze Prize (4th Place) — 2021
Nexon Youth Programming Challenge 2020 - Bronze Prize (8th Place) — 2020
Google Codejam 2021 - 196th Place (Round 3) — 2021
Google Kick Start 2020 - 28th Place (Round B) — 2020
Reply Code Challenge Teen Edition - 4th Place — 2021
Codeforces - Rating 2715 (International Grandmaster)

RESEARCH

Data Structure for Maintaining Kth Maximum / Minimum Element of Polynomial Functions — 2021

- Developed an efficient algorithm that can maintain Kth maximum / minimum element of linear functions, with geometric transformations using Modified Quad Tree (MQT) and Persistent Angular Sorting (PAS) data structure.
- Generalized PAS to arbitrary polynomial functions, and implemented MQT, PAS with buckets to show its efficiency.

Competitive Reinforcement Learning for Graph Coloring Games — 2020

- Devised a 2-player graph game in order to identify vertices important in graph connectivity, and found optimal strategies using reinforcement learning.
- Applied multi-agent reinforcement learning with DQN, DDQN, D3QN, and tested its efficiency in big graphs.

New Algorithm for Maintaining Lower/Upper Envelope of Similar Convex Figures — 2020

- Improved the Li-Chao Tree data structure to maintain the lower / upper envelope of quadratic functions.
- Proved that the improved algorithm can be applied to similar convex figures, and implemented it with buckets and divide & conquer methods to show its efficiency.

New Techniques for Dynamic Programming Optimization using Convex Hull and Segment Tree — 2019

- Implemented a dynamic convex hull data structure using splay trees, in order to optimize particular recurrences that can be reduced to finding extreme points in convex hull.
- Devised Quad Li-Chao Tree data structure and a bottom-up segment tree with convex hulls and tested its efficiency.

PROJECTS

Reinventing the Wheel Source code

- CircuitSim, a digital logic simulator that builds from gates and arithmetic circuits through memory systems to single-cycle and pipelined RV32I processors.
- Interactive circuit examples and component documentation.
- Timing models and experiments comparing processor implementations.

PS Checklist Source code

- A beta application for organizing programming contests, tracking problem progress, and recording practice sessions.
- ICPC and Olympiad catalogs, with Codeforces and AtCoder contest imports.
- Shared problem status, dated practice records, and a workflow for curating contest resources.

EXPERIENCE

Huawei Technologies Co., Ltd. Hong Kong Research Center, Research Intern 2025

Korean IOI Training Camp, Teaching Assistant 2021 - 2024

- Assisted in teaching various topics on algorithms to 50+ candidates of Korean IOI delegate

SKILLS

Languages C/C++ (Fluent), Java (Intermediate), Python (Intermediate), Scala (Basic)